

Giridhar Reddy Karnati

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RF and microwave instrumentation · wide-band antenna design and EM simulation · radio receiver characterisation · field-deployable ground stations for satellite telemetry.

EDUCATION

Sep 2022 – Jul 2026	B.E., Electronics & Communication Engineering; Minor in Computer Science Chaitanya Bharathi Institute of Technology (CBIT), Hyderabad ◦ CGPA 9.23 / 10 (as of May 2026) ◦ Academic Excellence Award (Silver Medal) for the highest CGPA in the ECE department, AY 2022–23 ◦ Coursework: Antennas & Wave Propagation, Microwave Engineering, RF & Microwave Circuits, Electromagnetic Theory, Communication Systems, Digital Signal Processing, Digital System Design (Verilog) ◦ Major project: <i>Design of Anti-Spoofing Antenna for Electronic Warfare</i> (guide: Dr. K. Suman)
2020 – 2022	Intermediate (Class XI–XII), Mathematics–Physics–Chemistry Sri Chaitanya Junior Kalasala, Hyderabad ◦ Aggregate: 96.2%
2020	Class X VijayaRatna High School

RESEARCH & PROFESSIONAL EXPERIENCE

Jul 2026 – present	NPTEL Pre-Doc Fellow (Project Associate) IC Design Lab, Dept. of Electrical Engineering, IIT Hyderabad · Advisor: Prof. Abhishek Kumar Front-end for a wideband receiver in CMOS.
May 2026 – Jul 2026	Facility Technologist Intern National Nanofabrication Centre (NNfC), CeNSE, IISc Bengaluru
Nov 2025 – May 2026	Visiting Research Student Astronomy & Astrophysics, Raman Research Institute (RRI), Bengaluru Architectural design and bring-up of a water-vapour radiometer; RF link-budget and thermal-noise analysis, and gain / noise-figure cascade modelling of the IF chain.
Aug 2025 – Sep 2025	Student Intern, Nanofabrication Lab (SPARC NANO-X) IIT Hyderabad Clean-room processing: laser lithography, sputtering, reactive-ion etching (RIE) and plasma cleaning. Trained on the IC design flow from layout through packaging; fabricated nano-silicon waveguides, microfluidic devices and 50 μm Au interdigitated electrodes for 2-D-material gas sensors using Heidelberg μPG 501 maskless lithography, DC magnetron sputtering and lift-off in a Class-1000 clean room.
Jul 2025 – Aug 2025	IEEE AP/MTT-S Summer Intern, RF & Microwave Circuits IIT Kharagpur (remote) · Mentor: Shrawan Kumar Patel RF and microwave circuit analysis and synthesis: designed and simulated microwave phase shifters and coupled-line couplers in Keysight ADS, with MATLAB models for comparison; evaluated insertion loss, phase error and return loss across the band.
Jul 2025	Electrical Engineering Summer School (EESS-25) Indian Institute of Science (IISc), Bengaluru Selected as one of 100 students nationwide for an intensive programme in signal processing, communications and microelectronics.

Jun 2025 – Jul 2025	Project Intern, SDR & Embedded Radio Systems National Institute of Technology Karnataka (NITK), Surathkal Built SDR-based transceivers and Raspberry Pi-driven portable APRS ground nodes for weather monitoring, deployable to remote field sites.
Mar 2025 – May 2026	Internship Trainee, Antenna Design & RF Hardware (part-time) RaoS Consultants LLC (remote) Comprehensive Internship Course on Antenna Design (CICAD 2025–26): electromagnetics, antenna theory, horns, reflectors and phased arrays, design and analysis, integration, and measurement. Tools: CST Microwave Studio. One of 24 students selected from 90+ applicants; 2nd Place Best Intern.
Dec 2024 – Jan 2025	Technical Intern India Space Week (remote)
Oct 2024	Research Trainee, ICAN 4.0 Workshop CREST, BITS Pilani Hyderabad Campus Fabricated and electrically characterised 2-D-material-based nano-device sensors.
Jun 2024 – Jul 2024	Internship Trainee National Institute of Electronics & Information Technology (NIELIT)

PUBLICATIONS

2025	VayuVani: A Self-Managed LoRa Ground Station for Amateur Operators K. Giridhar Reddy · HAM Fest India 2025 Magazine, Goa, pp. 20–27
2025	Design of Ground Station Antennas for Tracking of LoRa-Based Satellites at 433 MHz Amateur Frequencies K. Vasanth, K. Giridhar Reddy, V. Bala Sai Datta, K. Suman · Proc. IEEE SCEECs 2025, MANIT Bhopal · IEEE Xplore Best Research Paper Award
2024	Privacy-Preserving LoRa Ground Stations in Amateur Satellite Communications: A Case Study of VayuVani K. Giridhar Reddy · Radio Amateurs' World (RAW) Magazine · presented at HAM Fest India 2024, Kolkata

VayuVani is an open-source LoRa ground-station platform created by Chandra Tungathurthi (vayuvani.com). My articles and talks describe building and operating ground stations with it.

HONOURS & AWARDS

Aug 2026	IEEE SPACE Young Professional (YP) Laureate IEEE SPACE 2026 (Space, Aerospace and Defence Conference)
Jun 2026	Best Intern Award, 2nd Place CICAD 2025–26, RaoS Consultants LLC
2025 & 2026	GATE qualified, Electronics & Communication Engineering GATE 2025 (score 436) and GATE 2026
Jan 2026	Dayton Amateur Radio Association Scholarship ARRL Foundation, USA · featured in <i>QST</i> , ARRL's monthly journal
Dec 2025	B.Tech Initiative Travel Grant IEEE Microwaves, Antennas and Propagation Conference (MAPCON)
Dec 2025	Rank 20 of 250 (top 8%), AI/ML Powered FPGA Design & Simulation Hackathon National Institute of Technology Jamshedpur
Jul 2025	Selected for the IISc Electrical Engineering Summer School (EESS-25) Top 100 nationwide

Feb 2025	Best Research Paper Award IEEE SCEECs 2025, MANIT (NIT) Bhopal
Mar 2024	Academic Excellence Award (Silver Medal), AY 2022–23 Chaitanya Bharathi Institute of Technology · highest CGPA in the ECE department
2024	Three-time NPTEL Star Discipline Star (Computer Science & Engineering) and Believer Star, Jan–Apr 2024 · Enthusiast Star, Jul–Dec 2024
	2nd Prize, Technical Quiz, NAFED-08 Conference VSSC / ISRO, hosted at IIT Hyderabad
TALKS	
Oct 2025	VayuVani: your private LoRa satellite ground station HamFest India 2025, Goa College of Engineering, Farmagudi, Goa
Dec 2024	Privacy-preserving LoRa ground stations: a case study of VayuVani HAM Fest India 2024, Kolkata
LEADERSHIP & OUTREACH	
Oct 2023 – Apr 2026	Founder & Student President, then Senior Student Advisor HAM Radio Club of CBIT, Hyderabad - Founder & Student President, Oct 2023 – Nov 2025; Senior Student Advisor, Oct 2025 – Apr 2026. - Led technical projects and STEM-outreach workshops for school students. - Two-day workshop with the Dept. of ECE, “Design approaches of Antennas for HAM Radio Applications using Open Source Software” (Apr 2024): antenna modelling in 4NEC2, Morse code, cables and connectors, and licensing. - Hands-on HAM radio sessions for Class 10 students at ZPHS Nadergul.
2025	Technical Chair & Coordinator, SYNAPSE 2025 CBIT · managed all technical events for the college techno-cultural festival
PROFESSIONAL MEMBERSHIPS	
Jun 2025 – present	Member Applied Computational Electromagnetics Society (ACES)
May 2025 – present	Individual Member RISC-V International
Jan 2025 – present	Associate Member NoRCEL Institute (Network of Researchers on the Chemical Emergence of Life)
Nov 2024 – present	Student Member Institute of Electrical and Electronics Engineers (IEEE)
Sep 2024 – present	Student Member Radio Amateur Satellite Corporation (AMSAT)
Aug 2023 – Apr 2026	ISF Member (IETE Students' Forum) Institution of Electronics and Telecommunication Engineers (IETE)
LICENCES & CERTIFICATIONS	
Feb 2024	Amateur Wireless Station Licence, General Grade (callsign VU2TIO) Department of Telecommunications, Government of India
Sep 2026	IEEE MTT-S Virtual Workshop on AI for Microwaves IEEE Microwave Theory and Technology Society

Nov 2025, May 2026	IEEE Certificates Program (CEU/PDH), two certificates IEEE Educational Activities · topics include generative AI and 5G
Mar 2026	International Physics Competition (IPhC) 2025, semi-final round Participation certificate
Dec 2025	International Youth Math Challenge (IYMC) 2025, qualification round Participation certificate
Feb 2024	Certificate of Merit, Geo-data Sharing and Cyber Security Indian Institute of Remote Sensing (IIRS), ISRO

TECHNICAL SKILLS

EM & RF simulation	Ansyz HFSS, CST Studio Suite, Altair FEKO, Keysight ADS, MATLAB RF Toolbox
RF & microwave	Wide-band antennas, LNA, mixer, PLL/VCO, power amplifier, filter design, link-budget analysis, noise figure / IIP3, anechoic-chamber and VNA measurement workflows
RF hardware & bench	VNA, spectrum analyser, SDR (RTL-SDR, HackRF), soldering and RF connectorisation, PCB design and prototyping
Digital, VLSI & EDA	Cadence Virtuoso, Xilinx Vivado, Silvaco TCAD (Victory Process / Device / Extract), CleWin, SolidWorks
Programming & HDL	Python, C/C++, MATLAB, Verilog, VHDL, LabVIEW
Embedded	Arduino, Raspberry Pi, ESP32 (Xtensa LX6), RISC-V, Arm Cortex-M, FPGA prototyping
Fabrication	Photolithography, maskless laser writing, DC magnetron sputtering, reactive-ion etching (RIE), plasma cleaning, PVD, CVD, lift-off, RTA; I–V characterisation (Keithley 2450)
Languages	English (professional), Hindi (native / bilingual)